Aaron M. Bornstein **Updated: October 31, 2026**

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CURRENT POSITION

2025- University of California, Irvine
Associate Professor, Department of Cognitive Sciences
Department of Neurobiology and Behavior (by courtesy)
Center affiliations:
Center for the Neurobiology of Learning and Memory
Center for Theoretical Behavioral Sciences
Drury Institute for Translational Depression Discoveries
Center for Early Life Environment and Neuropsychiatric Outcomes

PREVIOUS APPOINTMENTS

2019-2025 Assistant Professor, Department of Cognitive Sciences, UC Irvine
2018 Associate Research Scholar, Princeton Neuroscience Institute
2013-2018 Postdoctoral Research Associate, Princeton Neuroscience Institute
2007-2013 Graduate researcher, New York University
2006 Research assistant, Deicken lab, UCSF/VA
2005-2007 Research assistant, Wagner lab, Stanford

EDUCATION

2013 Ph.D., Cognition & Perception, **New York University**
2003 S.B., Mathematics (additional concentration in Economics)
Massachusetts Institute of Technology

RESEARCH FUNDING

Ongoing

2026-2027 UCLA-CDU Dana Center Seed Grant (MPI with Carr, Jones, & Kalofonos)
"Critically examining neuroscience evidence for anosognosia as a transdiagnostic symptom"

2025-2030 NIMH R01MH061285 (subaward; PI SD Pollak)
"Emotion Processing: Risk for Psychopathology in Children"

2025-2030 NIA R01AG088306 (PI; MPI IJ Bennett)
"Memory-guided decision making across the lifespan"

2023-2028 NIMH R01MH128306 (Co-I; PI MA Yassa)
"Testing a memory-based hypothesis for anhedonia"

2022-2027 NINDS R01NS119468 (Co-I; PI ER Chrastil)
"Cognitive graphs, active decision making, and brain network dynamics"

Completed

2022-2025 NIMH P50MH096889 (Co-I; PI TZ Baram)
"Fragmented early-life experiences, aberrant circuit maturation, emotional vulnerabilities"

2022-2025 UK MRC MR/W028476/1 (Collaborator; PI M Field)
"Reinforcer-specific value-based decision-making in persistence of and recovery from alcohol use disorder"

2021-2025 NIA R21AG072673 (PI)
"Improving multi-step planning in aging by overcoming deficits in memory encoding"

2021-2023 BBRF NARSAD Young Investigator award (PI)
"Determining the role of episodic memory in substance use disorder"

2020-2021 NIMH P50MH096889 (seed grant; PI TZ Baram)
"Early life adversity effects on event segmentation"

2011-2013 NIMH F31MH095501 (PI)
"Computational mechanisms of goal-directed control"

To lab members/trainees

2027-2029 BBRF NARSAD Young Investigator award (Dale Zhou)
2026 UCI FIBRE (Neuroimaging center) project award (Rohin Palsule)
2026-2028 NIMH T32MH119049 (Alexa Booras)
2026 UROP Undergraduate Research Support Award (Dhriti Grover)
2026 UROP Undergraduate Research Support Award (Aidan Goeschel & Bryce Wilson)
2026 UROP Undergraduate Research Support Award (Annie Shen)
2025 NIA/SRNDNA Collaboration award (Shensheng Wang)
2025 UROP Undergraduate Research Support Award (Jianle Guo & Rohin Pasule)
2024 ARCS Foundation Scholar (Bianca Leonard)
2024 UCI Summer Undergraduate Research Fellowship (Meghan Johnson)
2024 Indow Fellowship for Research Excellence (Jungsun Yoo)
2024 NIA/SRNDNA Summer Research award (Melisa Azimihashemi)
2024 NIA/SRNDNA Summer Research award (Ami Yamamoto)
2024 UROP Research Experience Fellowship (Ami Yamamoto)
2023-2026 Hewitt Foundation postdoctoral fellowship (Dale Zhou)
2023-2025 NIMH F31MH134620 (Nora Harhen)
2023-2025 NIMH T32MH119049 (Ari Khoudary)
2023 Robert J. Glushko prize for best undergraduate thesis (Gloria Cheng)
2023 CNLM Jared M. Roberts Graduate Student Award (Ari Khoudary)
2023 CNLM Jared M. Roberts Graduate Student Award (Nidhi Banavar)
2022 UCI Summer Undergraduate Research Fellowship (Gloria Cheng)
2022 Lambert prize in Foundations of Science (Nidhi Banavar)
2022-2025 NIA F32AG072836 (Sharon Noh)
2020-2023 National Defense Science & Engineering Graduate fellowship (Nora Harhen)
2019 UCI Summer Undergraduate Research Fellowship (Brianna Sarcos)

Publications

(∗ = *Equal contribution*; + = *UCI Lab member.*)

Preprints

PP7 Copeland A, Stafford T, Crook C, Pun M, **Bornstein AM**, Hogarth L, Murphy JG, Pickard H, Ryan F, Strickland JC, Field M. The effect of alcohol-specific episodic memories and alcohol cue exposure on value-based decision-making and ad libitum alcohol consumption. *PsyArXiv*.
doi:10.31234/osf.io/3ka57
Submitted, Addiction

PP6 Khoudary A+, Peters MAK*, **Bornstein AM***. Memory retrieval explains dynamic effects of expectations on perceptual decisions. *bioRxiv*.
doi:10.64898/2026.09.22.753289
Under review, Nature Communications

PP5 Leonard BT+, Martínez-Ortíz MA, Bock J, Zhang Y+, Flores S, Taylor DV, Glynn LM, Davis EP, Stern HS, Baram TZ, Hartley CA, Yassa MA*, **Bornstein AM***. Anhedonia counteracts the effects of early-life unpredictability on threat-reward decision-making. *bioRxiv*.
doi:10.64898/2026.05.16.725643
Under review (2nd round), Proceedings of the National Academy of Sciences

PP4 Wang W, Chierchia G, Cooley BJ, Chang T, **Bornstein AM**, Schweizer S. Autobiographical Memory Specificity Shapes Navigation of Social Uncertainty in Depression. *PsyArXiv*.
doi:10.31234/osf.io/z6vej
Submitted, BMC Psychiatry

PP3 Yoo J+, **Bornstein AM**. Temporal dynamics of model-based control reveal arbitration between multiple task representations. *PsyArXiv*.
doi:10.31234/osf.io/sgcy5
Invited revision, Nature Communications Psychology

PP2 Zhang Y+*, Chen Y+*, Chen XR+, Harhen NC+, Glynn LA, Davis EP, Baram TZ, Risbrough VB, Stout DA, **Bornstein AM**. A specific computational role for early-life, and not lifelong, stress in decision-making under uncertainty. *bioRxiv*.
doi:10.64898/2026.06.01.729407
Submitted, Nature Human Behavior

PP1 Zhou D+, Noh SM+, Harhen NC+, Banavar NV+, Kirwan B, Yassa MA, **Bornstein AM**. A compressed code for memory discrimination. *bioRxiv*.
doi:10.1101/2025.10.12.681901
Under review (2nd round), Science Advances

Peer-reviewed journal articles

JP28 Harhen NC+, **Bornstein AM***, Hartley CA* (2026). Developmental changes in memory structure and precision alter the use of retrieved episodes during decisions for reward. *Proceedings of the National Academy of Sciences*, 123(31):e2525494123.
doi:10.1073/pnas.2525494123

JP27 Harhen NC+, Budiono R, Hartley CA*, **Bornstein AM*** (2026). Structure inference in complex environments improves from childhood to adulthood. *Developmental Science*, 29(3):e70163.
doi:10.1111/desc.70163

JP26 Noh SM+*, Zhou D+*, Cooper KW, Guo S+, Dinh ET+, **Bornstein AM** (2026). Sparsity and memory constraints interact with training sequence to bias learning of associative maps. *Neuropsychologia*, 225:109396.
doi:10.1016/j.neuropsychologia.2026.109396

JP25 Noh SM+*, Cooper KW*, Guo S+, Zhou D+, Stark CEL, **Bornstein AM** (2026). Multi-step inference can be improved across the human lifespan with individualized memory interventions. *The Journals of Gerontology, Series B: Psychological Sciences and Social Sciences*, 81(5):gbag038.
doi:10.1093/geronb/gbag038

JP24 Hadj-Amar B, **Bornstein AM**, Guindani M, Vannuci M (2025). Discrete Autoregressive Switching Processes with Cumulative Shrinkage Priors for Graphical Modeling of Time Series Data. *Journal of Computational and Graphical Statistics*.
doi:10.1080/10618600.2025.2551271

JP23 Schetzsle B, Lee J, **Bornstein AM**, Shahbaba B, Guindani M (2025). A Bayesian Time-Varying Psychophysiological Interaction Model. *Data Science in Science*, 4(1):2519436.
doi:10.1080/26941899.2025.2519436

JP22 Khoudary A+, Peters MAK*, **Bornstein AM*** (2025). Reasoning goals and representational decisions in computational cognitive neuroscience: lessons from the drift diffusion model. *European Journal of Neuroscience*, 61:e7009.
doi:10.1111/ejn.70098

JP21 Banavar NV+, Noh SM+, Wahlheim CN, Cassidy BS, Kirwan CB, Stark CEL, **Bornstein AM** (2024). A response time model of the three-choice Mnemonic Similarity Task provides stable, mechanistically interpretable individual-difference measures. *Frontiers in Human Neuroscience*, 18:137928.
doi:10.3389/fnhum.2024.137928

JP20 Yoo J+, Chrastil ER, **Bornstein AM** (2024). Cognitive graphs: Representational substrates for planning. *Decision*, 11(4), 537-556.
doi:10.1037/dec0000249

JP19 Chen J, **Bornstein AM** (2024). The causal structure and computational value of narratives. *Trends in Cognitive Sciences*, 28(8):769-781.
doi:10.1016/j.tics.2024.04.003

JP18 Harhen NC+, **Bornstein AM** (2024). Interval timing as a computational pathway from early life adversity to affective disorders. *Topics in Cognitive Science*, 16(2024):92-112.
doi:10.1111/tops.12701

JP17 Noh SM+, Singla UK, Bennett IJ, **Bornstein AM** (2023). Memory precision and age differentially predict the use of decision-making strategies across the lifespan. *Scientific Reports*, 13:17014.
doi:10.1038/s41598-023-44107-5

JP16 **Bornstein AM**, Aly M, Feng SF, Turk-Browne NB, Norman KA, Cohen JD (2023). Associative memory retrieval modulates upcoming perceptual decisions. *Cognitive, Affective, & Behavioral Neuroscience*, 23:645665.

doi:10.3758/s13415-023-01092-6
doi:10.18112/openneuro.ds001614.v1.0.1

JP15 Harhen NC+, **Bornstein AM** (2023). Overharvesting in human patch foraging reflects rational structure learning and adaptive planning. *Proceedings of the National Academy of Sciences*, 120(13):e2216524120.
doi:10.1073/pnas.2216524120

JP14 Otto AR, Devine S, Schultz E, **Bornstein AM***, Louie K* (2022). Context-dependent choice and evaluation in real-world consumer behavior. *Scientific Reports*, 12:17744.
doi:10.1038/s41598-022-22416-5
doi:10.17605/OSF.IO/EC5DX

JP13 Rmus M, Ritz H, Hunter LE, **Bornstein AM***, Shenhav A* (2022). Humans can navigate complex graph structures acquired during latent learning. *Cognition*, 225:105103.
doi:10.1016/j.cognition.2022.105103

JP12 Wang S, Feng SF, **Bornstein AM** (2021). Mixing memory and desire: How decisions for reward depend on the dynamics and content of memory reinstatement. *Wiley Interdisciplinary Reviews: Cognitive Science*, e1581.
doi:10.1002/wcs.1581
Featured on the cover.
Recognized as one of the top-ten most-cited papers from this journal/year.

JP11 Rouhani N, Norman KA, Niv Y, **Bornstein AM** (2020). Reward prediction errors create event boundaries in memory. *Cognition*, 203:104269.
doi:10.1016/j.cognition.2020.104269

JP10 **Bornstein AM***, Pickard H* (2020). Chasing the first high: Memory sampling in drug choice. *Neuropsychopharmacology*, 45(6):907-915.
doi:10.1038/s41386-019-0594-2 **Featured on the cover.**

JP9 Kane GA, **Bornstein AM**, Shenhav A, Wilson RC, Daw ND, Cohen JD (2019). Rats exhibit similar biases in foraging and intertemporal choice tasks. *eLife*, 8:e48429.
doi:10.7554/eLife.48429

JP8 Millner AJ, den Ouden HEM, Gershman SJ, Glenn CR, Kearns J, **Bornstein AM**, Marx BP, Keane TM, Nock MK (2019). Suicidal thoughts and behaviors are associated with an increased decision-making bias for active responses to escape aversive states. *Journal of Abnormal Psychology*, 128(2):106-118.
doi:10.1037/abn0000395

JP7 Hoskin AN, **Bornstein AM**, Norman KA, Cohen JD (2018). Refresh my memory: Episodic memory reinstatements intrude on working memory maintenance. *Cognitive, Affective, & Behavioral Neuroscience*, 19:338-354.
doi:10.3758/s13415-018-00674-z
doi:10.18112/openneuro.ds001576.v1.0.0

JP6 **Bornstein AM**, Khaw MW, Shohamy D, Daw ND (2017). Reminders of past choices bias decisions for reward in humans. *Nature Communications*, 8:15958.
doi:10.1038/ncomms15958

JP5 **Bornstein AM**, Norman KA (2017). Reinstated episodic context guides sampling-based decisions for reward. *Nature Neuroscience*, 20:997-1003.
doi:10.1038/nn.4573
doi:10.18112/openneuro.ds001607.v1.0.1

JP4 **Bornstein AM**, Daw ND (2013). Cortical and hippocampal correlates of deliberation during model-based decisions for rewards in humans. *PLoS Computational Biology*, 9(12):e1003387.
doi:10.1371/journal.pcbi.1003387

JP3 **Bornstein AM**, Daw ND (2012). Dissociating hippocampal and striatal contributions to sequential prediction learning. *European Journal of Neuroscience*, 35:1011-1023.
doi:10.1111/j.1460-9568.2011.07920.x

JP2 **Bornstein AM**, Daw ND (2011). Multiplicity of control in the basal ganglia: computational roles of striatal subregions. *Current Opinion in Neurobiology*, 21(3):374-380.
doi:10.1016/j.conb.2011.02.009

JP1 Preston AR, **Bornstein AM**, Hutchinson JB, Gaare ME, Glover GH, Wagner AD (2010). High-resolution fMRI of content-sensitive subsequent memory responses in human medial temporal lobe. *Journal of Cognitive Neuroscience*, 22:156-173.
doi:10.1162/jocn.2009.21195

Articles in conference proceedings

CP10 Zhou D, Irizarry-Mártinez G, Guo S, Kim S, Janecek J, Tuteja N, Meza N, McMillan L, Thayer J, **Bornstein AM***, Yassa MA* (2026). Effects of surprising rewards on pattern separation. *Proceedings of the 48th Annual Conference of the Cognitive Science Society.*

CP9 Khoudary A+, **Bornstein AM***, Peters MAK* (2025). Investigating implicit and explicit expectations in perceptual decision making. *Proceedings of the 47th Annual Conference of the Cognitive Science Society.*

CP8 Banavar NV+, **Bornstein AM** (2022). Response time modeling provides stable and mechanistically interpretable measures of individual differences in behavioral pattern separation. *Proceedings of the 20th International Conference on Cognitive Modeling.*
Selected for a talk.

CP7 Harhen NC+, **Bornstein AM** (2022). Learning to expect change: Volatility during early experience alters reward expectations in a model of interval timing. *Proceedings of the 20th International Conference on Cognitive Modeling.*
Selected for a talk.
Selected for a best paper award.

CP6 Banavar NV+, **Bornstein AM** (2022). Decision difficulty modulates the re-use of computations across trials in non-sequential decision tasks. *Proceedings of the 5th Multidisciplinary Conference on Reinforcement Learning and Decision Making (RLDM 2022)*

CP5 Banavar NV+, Lee MD, **Bornstein AM** (2021). Sequential effects in non-sequential tasks. *Proceedings of the 19th International Conference on Cognitive Modeling.*

CP4 Harhen NC+, **Bornstein AM** (2021). Structure learning as a mechanism of overharvesting. *Proceedings of the 19th International Conference on Cognitive Modeling.*

CP3 Harhen NC+, Hartley CA, **Bornstein AM** (2021). Model-based foraging using latent-cause inference. *Proceedings of the 43rd Annual Conference of the Cognitive Science Society.*
doi:10.31234/osf.io/dfztu
https://github.com/uciccnl/CogSci2021-HarhenHartleyBornstein

CP2 Kane GA, **Bornstein AM**, Shenhav A, Wilson RC, Daw ND, Cohen JD (2017). Mechanisms of overharvesting in patch foraging. *Proceedings of the 39th Annual Conference of the Cognitive Science Society*, 637-642.

CP1 Floares A, Jakary A, **Bornstein A**, Deicken R (2006). Neural networks and classification and regression trees are able to distinguish females with major depression from healthy controls using neuroimaging data. *Proceedings of the IEEE International Joint Conference of Neural Networks, 2006*, 4605-4611.
doi:10.1109/ijcnn.2006.247090

Commentaries & Book Chapters

RP8 Zhou D+, **Bornstein AM** (2026). Decompressing the kinds of compression (Commentary on Dubova & Sloman). *Behavioral and Brain Sciences*.
doi:10.1017/S0140525X26107511

RP7 Khoudary A+, **Bornstein AM***, Peters MAK* (2026). Blind ignorance or rational inattention? On the history and future of metabolic considerations for cognitive models (Commentary on Haueis & Colaco). *Behavioral and Brain Sciences*, 49, e466.
doi:10.1017/S0140525X26106293

RP6 Zhou D+, **Bornstein AM** (2024). Expanding horizons in reinforcement learning for curious exploration and creative planning (Commentary on Ivancovsky et al). *Behavioral and Brain Sciences*, 47, e118.
doi:10.1017/S0140525X23003394

RP5 Banavar NV+, **Bornstein AM** (2024). Multi-plasticities: Distinguishing context-specific habits from complex perseverations. In Y. Vandaele (Ed.). *Habits - Their Definition, Neurobiology and Role in Addiction*. Cham, Switzerland: Springer Nature.

RP4 **Bornstein AM***, Constantino SM* (2017). Nudge back: Towards a taxonomy of scientific rationalities. *London Conference in Critical Thought*.

RP3 **Bornstein AM**, Miller KJ, Shenhav A (2015). Walking bundles of habits (and Response-Outcome associations). *European Journal of Neuroscience*, 41:1356-1357.
doi:10.1111/ejn.12906

RP2 **Bornstein AM** (2014). Functions of the hippocampal memory system in instrumental control (Doctoral dissertation).
Available from ProQuest Dissertations & Theses Global (3614853).

RP1 **Bornstein AM**, Nylen EL, Steele SA (2011). Unblocking the neural substrates of model-based value. *Journal of Neuroscience*, 31(28):10117-10118.
doi:10.1523/jneurosci.1883-11.2011

Accepted conference abstracts (since 2019)

CA82 Maxwell M, Wang E, Palsule R, Chen XR, Johnson MP, Harhen NC, Noh SM, **Bornstein AM***, Bennett IJ* (2026). Age Group Differences and Structural Substrates of Structured Foraging. *Scientific Research Network on Decision Neuroscience and Aging*

CA81 Noh SM+, Morton NW, **Bornstein AM**, Preston AR (2026). Interleaving supports category generalization by integrating exemplars and differentiating categories. *Psychonomics*.
Selected for a talk.

CA80 Noh SM+, Morton NW, **Bornstein AM**, Preston AR (2026). Interleaving learning shapes neural representations in medial prefrontal cortex to enhance real-world category knowledge.
Nanosymposium: Behavioral and Neural Insights Into Representational Learning.
Society for Neuroscience Annual Meeting.

CA79 Guo S+, Zhou D+, Chen XR+, **Bornstein AM** (2026). Memory-regularized variational autoencoders capture training sequence effects and develop predictive sparse representations connecting to a successor representation.
Nanosymposium: Behavioral and Neural Insights Into Representational Learning.
Society for Neuroscience Annual Meeting.

CA78 Cooper KW+, Noh SM+, Stark CEL, Fortin NJ, **Bornstein AM** (2026). The order of experience shapes multi-step representations through multi-step decisions across multiple species.
Nanosymposium: Behavioral and Neural Insights Into Representational Learning.
Society for Neuroscience Annual Meeting.

CA77 Grover D+, Leonard BT+, **Bornstein AM** (2026). Reaction Time and Behavioral Avoidance in Anxiety: Evidence from Human Prey Foraging. *Society for Neuroeconomics Annual Meeting.*

CA76 Shen A+, **Bornstein AM** (2026). Joint Modeling of Response Time and Choice Reveals Asymmetric Switch Costs Under Risk and Ambiguity. *Society for Neuroeconomics Annual Meeting.*

CA75 Zhou D+, Guo S+, Yassa MA, **Bornstein AM** (2026). Representation learning and choice in naturalistically complex environments. *Society for Neuroeconomics Annual Meeting.*

CA74 Palsule R+, Goeschel A+, Yoo J+, Chen J, Khoudary A, **Bornstein AM** (2026). The relationship between contextual information gain and default mode network activity during narrative encoding. *Bay Area Memory Meeting*
Selected for a talk.

CA73 Cooper KW+, Noh SM+, Stark CEL, Fortin NJ, **Bornstein AM** (2026). The order of things: mnemonic discrimination, capacity, and replay shape multi-step inference. *Bay Area Memory Meeting*
Selected for a talk.

CA72 Zhou D+, Harhen NC+, Davis EP, Glynn LM, Baram TZ, Risbrough VB, Stout DM, Yassa MA*, **Bornstein AM*** (2026). Early-Life Unpredictability and the Neural and Computational Timescales of Integration. *The Flux Society Congress for Integrative Developmental Neuroscience.*
Selected for a talk.

CA71 Booras AK, Kapogianis TK, Yoo J, **Bornstein AM***, Chrastil ER* (2026). Individual Variation in Non-Spatial Navigation is Related to Adaptive Learning and Use of Successor Representations. *Cognitive Computational Neuroscience.*

CA70 Goeschel A*, Palsule R*, Yoo J, Chen J, **Bornstein AM**, Khoudary A (2026). Quantifying information gain in narrative stimuli using open-source large language models. *Cognitive Computational Neuroscience.*

CA69 Guo S*, Booras AK*, Kahana MJ, Fried I, Sakon J, **Bornstein AM** (2026). Replay Supports Both Future and Immediate Decisions by Updating a Successor Representation. *Cognitive Computational Neuroscience.*

CA68 Chen XR*, Johnson MJ*, Palsule RA, Maxwell M, Harhen NC, Noh SM, Bennett IJ, **Bornstein AM** (2026). Age-related differences in structure learning drive foraging behavior in a multimodal patch-leaving task. *Cognitive Computational Neuroscience.*

CA67 Zhou D, Guo S, Yassa MA, **Bornstein AM** (2026). Representation learning and choice in naturalistically complex environments. *Cognitive Computational Neuroscience.*

CA66 Zhang Y, **Bornstein AM** (2026). An adaptive role for anhedonia following early-life unpredictability. *Cognitive Computational Neuroscience*.

CA65 Zhang Y, **Bornstein AM** (2026). Belief-space warping as an adaptive response to early-life unpredictability. *Computational Psychiatry 2026*.

CA64 Zhou D+, Harhen NC+, Stout DM, Jirsaraie R, Parker D, Risbrough V, Davis EP, Glynn LM, Baram TZ, Yassa MA*, **Bornstein AM*** (2026). Adaptation to early life unpredictability: elongated temporal representations for reward-based decision making. *Computational Psychiatry 2026*.
Selected for a short talk.

CA63 Booras AK+, **Bornstein AM***, Chrastil ER* (2026). Non-spatial navigation is driven by both adaptive learning and use of successor representations. *CNLM Annual Meeting*.
Selected for a short talk.

CA62 Guo S+, Booras AK+, Sakon J, **Bornstein AM** (2026). Replay supports future decisions by updating a successor representation. *CNLM Annual Meeting*.
Selected for a short talk.

CA61 Palsule R+, Goeschel A+, Yoo J+, Chen J, **Bornstein AM** (2026). Comparing a large language model-derived metric of contextual information gain with human judgements in narratives. *CNLM Annual Meeting*.

CA60 Zhou D+, Harhen NC+, Davis EP, Glynn LM, Baram TZ, Risbrough VB, Stout DM, Yassa MA*, **Bornstein AM*** (2026). Early-Life Unpredictability and the Neural and Computational Timescales of Integration. *Anxiety and Depression Association Annual Meeting*.

CA59 Leonard BT+, Martínez-Ortiz MA, Hartley CA, Yassa MA*, **Bornstein AM*** (2026). Anhedonia and early life unpredictability distinctly shape motivational conflict resolution in adults. *American College of Neuropsychopharmacology*.

CA58 Booras A+, Chrastil ER*, **Bornstein AM*** (2025). Humans adaptively discount successor representations when navigating graph-structured spaces. *Bay Area Memory Meeting*.

CA57 Khoudary A+, Peters MAK*, **Bornstein AM*** (2025). Uncertainty resolution explains dynamic effects of memory on perceptual decisions *Bay Area Memory Meeting*.
Selected for a talk.

CA56 Zhou D+, Yassa MA*, **Bornstein AM*** (2025). A lossy compression account of mnemonic discrimination: Computational theory and multimodal evidence. *Bay Area Memory Meeting*.
Selected for a talk.

CA55 Booras A+, Chrastil ER, **Bornstein AM** (2025). Humans adaptively discount successor representations when navigating graph-structured spaces. *Society for Neuroscience Annual Meeting*.

CA54 Guo S+*, Palsule R+*, Noh SM+*, Preston AR, **Bornstein AM** (2025). The impact of learning sequences in encouraging distinct memory representations to support associative inference in older-adults. *Society for Neuroscience Annual Meeting.*

CA53 Khoudary A+, **Bornstein AM***, Peters MAK* (2025). A philosophical toolkit for computational modeling of cognition. *Society for Neuroscience Annual Meeting.*

CA52 Khoudary A+, Peters MAK*, **Bornstein AM*** (2025). Precision-weighted integration explains dynamic effects of expectations on perceptual decisions. *Society for Neuroscience Annual Meeting.*

CA51 Leonard BT+, Martínez-Ortíz MA, Hartley CA, Yassa MA, **Bornstein AM** (2025). Motivational conflict resolution is shaped by early life unpredictability. *Society for Neuroscience Annual Meeting.*

CA50 Palsule R+*, Guo S+*, Noh SM+*, **Bornstein AM** (2025). Naturalistic task framing improves older adults' ability to infer and navigate complex associative networks. *Society for Neuroscience Annual Meeting.*

CA49 Yoo J+, Zhou D+, Goeschel A+, **Bornstein AM** (2025). Contingency-dependent state augmentation as a normative learning rule for non-Markovian tasks. *Society for Neuroscience Annual Meeting.*

CA48 Field M, Copeland A, Hogarth L, Pun M, Ryan F, Pickard H, **Bornstein AM**, Murphy J, Strickland J (2025). Meaning in life and value-based decision-making in alcohol use disorder. *55th EABCT Congress.*

CA47 Zhou D*, Noh SM*, Cooper KW, Guo S, Dinh ET, **Bornstein AM** (2025). Sparse and distributed memory constraints drive representational change as a function of temporal learning sequence. *47th Annual Conference of the Cognitive Science Society.*

CA46 Khoudary A+, Peters MAK*, **Bornstein AM*** (2025). A principled model of uncertainty resolution explains dynamic effects of expectations in perceptual decision making. *Society for Philosophy and Neuroscience 2025.*
Selected for a talk.

CA45 Chen Y+, Harhen NC+, Stout DA, **Bornstein AM** (2024). Early life unpredictability modulates planning horizon in a structured foraging task. *Cognitive Computational Neuroscience.*

CA44 Harhen NC+, Hartley CA*, **Bornstein AM*** (2024). Development of structure inference contributes to age-related differences in exploration. *Cognitive Computational Neuroscience.*

CA43 Yoo J+, Zhou D+, **Bornstein AM** (2024). Latent cause inference as an efficient and flexible learning rule for cognitive graphs. *Cognitive Computational Neuroscience.*

CA42 Zhou D+, Noh SM+, Yassa MA, **Bornstein AM** (2024). Pattern separation using compressed and semantic representations of memory. *Cognitive Computational Neuroscience.*

CA41 Khoudary A+, Peters MAK*, **Bornstein AM*** (2022). Precision-weighted evidence integration predicts time-varying influence of memory on perceptual decisions. *Cognitive Computational Neuroscience.*

CA40 Kapogianis T, **Bornstein AM**, Chrastil ER (2024). Size and community structure affect abstract graph learning. *45th Annual Conference of the Cognitive Science Society.*

CA39 Azimihashemi M+, Banavar NV+, Guo S+, **Bornstein AM** (2024). Does alcohol dependence differentially influence episodic and semantic contributions to recognition priming in older adults? *SRNDNA Summer Symposium 2024.*

CA38 Yamamoto A+, Guo S+, Nadkarni M+, Rouhani N, **Bornstein AM** (2024). Alcohol Prediction Errors in Individuals With Varying Levels of Alcohol Use. *SRNDNA Summer Symposium 2024.*

CA37 Dinh ET+, Zhou D+, Guo S+, Noh SM+, Cooper KW, **Bornstein AM** (2024). Autoencoder models of human graph learning reveal that sparse and dense representations differentially support planning and recall *Society for Neuroscience Annual Meeting.*

CA36 Harhen NC+, **Bornstein AM***, Hartley CA* (2024). Reinstated episodes and context differentially guide decision making across development. *Society for Neuroscience Annual Meeting.*

CA35 Chen Y+, Harhen NC+, Stout DM, **Bornstein AM** (2024). Early-life unpredictability modulates planning horizon in a structured foraging task. *Society for Neuroscience Annual Meeting.*

CA34 Yoo J+, **Bornstein AM** (2024). Goal-directed control evolves in tandem with multiple task representations. *Society for Neuroscience Annual Meeting.*

CA33 Khoudary A+, **Bornstein AM***, Peters MAK* (2024). Reasoning goals and representational decisions in computational cognitive neuroscience: a case study from bounded evidence accumulation. *Philosophy and Neuroscience at the Gulf VII.*

CA32 Martínez-Ortiz MA, Leonard B+, Yassa MA, **Bornstein AM**, Hartley CA (2024). Exploring the relationship between Early Life Adversity (ELA) dimensions and the development of decision-making strategies under motivational conflict. *The Flux Society Congress for Integrative Developmental Neuroscience.*

CA31 Khoudary A+, Peters MAK*, **Bornstein AM*** (2024). Explaining dynamic effects of memory on perceptual decisions. *Learning and Memory 2024.*

CA30 Khoudary A, **Bornstein AM***, Peters MAK* (2023). Characterizing dynamic effects of memory on perceptual evidence accumulation. *RIKEN CBS Summer Program*

CA29 Chen Y+, Yoo J+,Harhen NC+, Noh SM+, Stout DM*, **Bornstein AM*** (2023). Adult and early-life adversity influences on memory-guided planning. *UCI Conte Center annual symposium.*

CA28 Harhen NC+, Noh SM+, Stough-Lacking S+, **Bornstein AM** (2023). Suboptimal or locally rational? Foraging as a window onto the universal mechanisms of decisions under uncertainty. *Healing the Brain: Bridging the Gap in Low-and Middle-Income Countries.*

CA27 Harhen NC+, **Bornstein AM** (2023). Temporal representation optimization provides a computational link between early life adversity and anhedonia. *Computational Psychiatry 2023.*

CA26 Banavar NV+, **Bornstein AM** (2023). Decomposing behavioral pattern separation: A model-based analysis. *Learning and Memory 2023.*

CA25 Harhen NC+, **Bornstein AM** (2023). Temporal representation adaptation as a computational link between early life unpredictability and anhedonia. *Learning and Memory 2023.*

CA24 Khoudary A+, Peters MAK*, **Bornstein AM*** (2023). Characterizing dynamic effects of memory on perceptual decisions. *Learning and Memory 2023.*

CA23 Noh SM+, **Bornstein AM** (2023). Memory encoding ability interacts with training interventions to improve memory-guided inference decisions *Learning and Memory 2023.*

CA22 Yoo J+, **Bornstein AM** (2023). Humans build configural representations for planning in complex environments *Learning and Memory 2023.*

CA21 Stout DM, Harhen NC+, **Bornstein AM**, Vinograd M, Spadoni A, Simmons AN, Yassa MA, Davis EP, Glynn LM, Baram TZ, Baker DG, Risbrough VB (2023). Unpredictable early-life experiences moderate the effect of anhedonia and PTSD symptoms on neural measures of reward learning in adulthood. *Anxiety and Depression Association of America.*

CA20 Harhen NC+, **Bornstein AM** (2023). Interval timing as a computational framework for examining the pathway from early life unpredictability to affective disorders. *UCI Conte Center annual symposium.*
Best abstract award.
Selected for a short talk.

CA19 Khoudary A+, **Bornstein AM***, Peters MAK* (2023). Perceptual decisions result from dynamic precision-weighted integration of memory and visual information. *Association of Scientific Studies of Consciousness.*
Selected for a talk.

CA18 Kapogianis T, **Bornstein AM**, Chrastil ER (2022). Graph Metrics and Non-Spatial Navigational Learning. *Society for Neuroscience Annual Meeting.*

CA17 Harhen NC+, Hartley CA*, **Bornstein AM*** (2022). Memory-guided decision-making develops alongside model-based planning. *The Flux Society Congress for Integrative Developmental Neuroscience.*

CA16 Harhen NC+, **Bornstein AM** (2022). Humans adapt their foraging strategies and computations to environment complexity. *Proceedings of the 5th Multidisciplinary Conference on Reinforcement Learning and Decision Making (RLDM 2022)*

CA15 Yoo J+, **Bornstein AM** (2022). Two-stage task with increased state space complexity to assess online planning. *Proceedings of the 5th Multidisciplinary Conference on Reinforcement Learning and Decision Making (RLDM 2022)*

CA14 Harhen NC+, **Bornstein AM** (2022). Representation learning & adaptation in human foraging. *CNLM annual meeting.*
Selected for a short talk.

CA13 Yoo J+, **Bornstein AM** (2022). Task complexity and experience modulate the use of online planning. *CNLM annual meeting.*

CA12 Noh SM+, Stark CEL, **Bornstein AM** (2022). Mnemonic Discrimination Ability Predicts Optimal Training Condition for Memory-Guided Inference Decisions. *Annual meeting of the Cognitive Neuroscience Society.*
Selected for a short talk.

CA11 Yoo J+, **Bornstein AM** (2021). Task complexity and experience dictate the use of online, versus offline, planning in humans. *Annual meeting of the Society for Neuroeconomics.*

CA10 Noh SM+, Kerr T, Bennett IA*, **Bornstein AM*** (2021). Age-related differences in memory-guided decisions are driven by a trade-off between multiple decision systems. *Society for Neuroscience Annual Meeting.*

CA9 Noh SM+, Kerr T, **Bornstein AM** (2021). Pattern Separation Predicts Which Memories Are Sampled During Decisions for Reward *Psychonomics.*

CA8 Noh SM+, **Bornstein AM** (2021). Pattern separation mediates the types of memories sampled during decisions for reward. *CNLM annual meeting.*

CA7 Banavar NV+, **Bornstein AM** (2021). Deliberative evaluation in intertemporal choice is shaped by experiment structure. *Annual meeting of the Society for Neuroeconomics*
Selected for a short talk.

CA6 Devine SM, Otto AR, **Bornstein AM***, Louie K* (2021). Context-dependent choice and evaluation in real-world consumer behavior *Mathematical Psychology*

CA5 Harhen NC+, Yassa MA, Baram TZ, **Bornstein AM** (2021). Exploring a latent cause model of substance use disorder symptoms. *Biological Psychiatry* doi:10.1016/j.biopsych.2021.02.479

CA4 Cooper KW+, Li L, Agostinelli F, Saraf M, Elias GA, Baldi P, **Bornstein AM**, Shahbaba B, Fortin N (2021). Theta-associated nonspatial sequence coding in hippocampus. *Society for Neuroscience Global Connectome.*

CA3 Harhen NC+, Hartley CA, **Bornstein AM** (2020). Foraging behavior adjusts to multiple scales of context. *Annual meeting of the Society for Neuroeconomics* **Selected for a talk.**

CA2 Vlasceanu M, Morais MJ, Zhao Z, **Bornstein AM**, Norman KA, Coman AC (2020). Self-Other Similarity Modulates the Socially-Triggered Context-Based Prediction Error Effect on Memory. *41st Annual Conference of the Cognitive Science Society*

CA1 Rouhani N, Norman KA, Niv Y, **Bornstein AM** (2019). Reward prediction errors create event boundaries in memory *Psychonomics*

Awards

2024	CNLM Exceptional Mentor Award
2023	Election to the Memory Disorders Research Society
2020	Association for Psychological Science "Rising Star" Award
2020	Brain and Behavior Research Foundation NARSAD Young Investigator Award
2005,6,8	Honorable mention, NSF Graduate Research Fellowship
2000	Runner up, MIT $50k Business Plan competition

Talks

Invited

Nov 2026	CalState Long Beach
Nov 2025	Mechanistic Basis of Foraging, Birmingham, UK
Mar 2025	Rutgers-Princeton Center for Computational Cognitive Neuro-Psychiatry
Feb 2025	University of California, Berkeley
Feb 2025	National Institute of Drug Abuse
Nov 2024	Singidunum University, Belgrade, Serbia
Sep 2024	New York University
Feb 2024	University of California Los Angeles
Feb 2024	University of Miami
Jan 2024	Virginia Tech
Dec 2023	University of California Davis
Dec 2023	Brown University
May 2023	Future of Foraging seminar series
Apr 2023	Johns Hopkins University

Mar 2023	Northeastern University
Feb 2023	Center for the Neurobiology of Learning & Memory
Dec 2022	City University of New York
Feb 2021	University of California Los Angeles
Feb 2021	Context and Affective Memory meeting
Aug 2020	University of California Los Angeles
Mar 2020	Claremont Colleges
Feb 2020	University of California Riverside
Sep 2019	Facebook Research Labs
May 2019	University of California Los Angeles
Apr 2019	Brown University
Jan 2019	National Institute of Drug Abuse
Dec 2017	Cambridge University
Oct 2017	Johns Hopkins University
Feb 2015	Mount Sinai School of Medicine
Jun 2014	Weill-Cornell Medical College
Jun 2011	Memory and Decisions meeting, Stanford University
Jan 2011	Parallel Distributed Processing meeting, Princeton University
Jan 2011	Kavli Institute, Harvard University

Contributed

Apr 2025 Cognitive Neuroscience Society symposium, "Uncertainty Resolution across Learning, Memory, and Decision-making"

Jan 2025 Workshop on Self-assembling Games, UC Irvine

May 2024 American Psychological Society (Presenter: +Sharon M Noh)

May 2022 Center for the Neurobiology of Learning and Memory Annual Meeting

Apr 2021 Society of Biological Psychiatry symposium "Advancing Treatment of Substance Use Disorders Through Computational Modeling"

Jan 2020 Winter Conference on the Neurobiology of Learning and Memory symposium "Mnemonic contributions to reinforcement learning and decision making"

Mar 2018 Cosyne workshop "Hippocampal computations and interactions supporting statistical learning and decision-making"

Nov 2015 SFN symposium "Understanding Goal-Directed Decision-Making in Humans: Computations and Circuits"

Mar 2014 Workshop on the Neurobiology of Prediction and Surprise, Rutgers University

Teaching/Service

UC Irvine

Spring 2027	Neural Data Analysis (Undergraduate)
Fall 2026	Cognitive Maps (Graduate; cross-listed with Critical Theory Emphasis)
Fall 2025	Resisting "Neuro-narratives" (Undergraduate)
Winter 2025,2027	LIFTED UC Irvine Prison Education Program (Human Memory)
Spring 2023-2026,Winter 2027	Human Memory (Undergraduate)
Spring 2023-24,Fall 2025	Decision making (Graduate)
Spring 2021-22	Decision making & Problem solving (Graduate; with Prof. Zygmunt Pizlo)
Spring 2020	Topics in Reinforcement Learning (Graduate; with Prof. Mimi Liljeholm)
Spring 2019-22	Research in Exp Psych (Undergraduate; with Prof. Nadia Chernyak)
Winter 2019-22	Advanced Experimental Psychology (Undergraduate)

Other teaching/mentorship

Fall 2020	Neuromatch academy mentor
Summer 2020	Neuromatch academy group leader
Summer 2020	GSMI Cientifico Latino
2019	MEET alumni mentor
Summer 2018	Computational & Cognitive Neuroscience Summer School, Suzhou, China
2016-2018	Princeton Prison Teaching Initiative
Summer 2007,08	MIT Middle East Education through Technology (MEET), Jerusalem

University/Department service

2024-	UCI Research Imaging Equipment Committee
2022-2025	Cognitive Sciences Colloquium committee, faculty advisor
2021-	Irvine Faculty Association, Executive board member; 2023-2025 Treasurer; 2025- Co-Chair
2020	CNLM "Evening to Remember" organizational committee
2019-	First-Generation Faculty Initiative
2019-	UCI Prison Education Program, advisory board, curriculum committee
2019,23	Cognitive Sciences faculty search committee
2018,20,22,24	Cognitive Sciences PhD admissions committee

Organizational/Field service

2025-	Associate Editor, *Cognitive Science*.
2025-	Justice, Diversity, Equity, and Inclusion Committee, *Memory Disorders Research Society*.
2025-	Advisor, *All People's Health Collective*.
2024-	Editorial board, *Translational Neuroscience*.
2023	UCI Conte Center annual symposium, organized with the Conte Center Team.

2022 NSF/Simons NeuroDataScience workshop, Co-Organizer (with Norbert Fortin and Babak Shahbaba)

2021 Center for Neurobiology of Learning and Memory Spring Meeting Co-organizer (with Lulu Chen)

2018 Princeton Neuroscience Institute "Inside-Out" seminar series Co-organizer (with Ahmed El Hady)

2018 "Goal-Directed Decision Making: Computations and Circuits" *Elsevier* Co-editor (with Richard Morris & Amitai Shenhav)

2015 COSYNE Workshop "Memory in action: The role(s) of the hippocampus in decisions for reward" Co-organizer (with G. Elliott Wimmer)